

Rasmus Frigaard Lemvig - Curriculum Vitae

Birthday: 1 January, 1999

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Homepage: <https://rasmusfl.github.io/>

Education

University of Copenhagen

PhD student. Principal supervisor: Martin Rainer Bladt. Co-supervisor: Mogens Steffensen (*May 2025 - present*)

Working thesis title: *Multi-state models: modern methods, machine learning and meditations* ((MM)³)

University of Copenhagen

Master of Science in Actuarial Mathematics (*Sep 2023 - Jan 2025*)

Thesis: Machine learning methods for survival and multi-state models - Supervised by Martin Rainer Bladt

University of Copenhagen

Bachelor of Science in Actuarial Mathematics (*Sep 2021 - Jun 2023*)

University of Copenhagen

Bachelor of Science in Mathematics (*Sep 2018 - Jun 2021*)

Thesis: Cubic and Quartic Reciprocity - Supervised by Ian Kiming

Experience

University of Copenhagen - Department of Mathematical Sciences

PhD student (*May 2025 - present*). PhD student in the section of Insurance and Economics (IE). TA in the courses Statistics for Insurance and Quantitative Risk Management. Primarily working with estimation in multi-state models.

University of Copenhagen - Department of Mathematical Sciences

Scientific assistant (*Feb 2025 - May 2025*). TA in the courses Quantitative Risk Management and Topics in Life Insurance. Working on a research project with Martin Bladt.

Teaching assistant (*Sep 2020 - Jan 2025*) in 17 different courses overall. Noteworthy courses include:

- Topics in Probability
- Basic Life Insurance Mathematics
- Statistical Methods
- Brownian Motion
- Continuous Time Finance
- Advanced Probability Theory 2
- Stochastic Processes in Non-Life Insurance
- Insurance and Law
- Probability Theory 2

- Analysis 0
- Probability Theory
- Lebesgue Integral and Measure Theory

My complete teaching history can be found here: <https://rasmusfl.github.io/work.html>.

GoTutor

Freelance Tutor (2019-2020)

Tutoring in high school mathematics.

Self-employed

Tutor (Sep 2021 - present)

Tutoring in various courses at the University of Copenhagen.

Manuscripts and projects

1. *Consistency of Honest Decision Trees and Random Forests* with Martin Bladt.
 2. *Local estimation of transition rates of jump processes through discretization* with Martin Bladt.
 3. *JumpForests: A random forest R package for jump processes* with Martin Bladt.
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Academic interests

The mathematics of life insurance, in particular multi-state modelling. Machine learning and statistics related to life insurance. Mathematical aspects of risk management and mathematical finance.

Programming proficiencies

Experience with programming in **R**, **Julia**, **Python/Cython** and **C++**.

Lecture notes

I have written lecture notes for the following courses (available on my homepage):

- *Quantitative Risk Management (QRM) 2025/2026*. Currently working on making these notes into a book with Jeffrey Collamore.
 - *Stochastic Processes in Non-Life Insurance (SkadeStok) 2023/2024*
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Volunteer work

Danish Youth Association of Science (UNF)

Volunteer (Jan 2016 - present). Some highlights include:

- **Cofounder of, coordinator and teacher at UNF Science Weekend** (2019 - 2023), a biannual weekend camp for elementary school students with 85 participants and 30 volunteers.
- **Teacher at Mathematics Camp** from 2021 to 2025 and at Machine Learning Camp 2024.
- **Teacher at Week in Mathematical Modelling 2025**, a camp with focus on differential equations and their applications. Teacher and coordinator in 2026 also.

- **Teacher** in several workshops. Topics: Linear algebra, calculus, number theory, graph theory, sums and sequences, differential equations.
- **Coordinator of and teacher in** the project Evenings of Natural Science Education where we taught young Ukrainian refugees.

A detailed description of my work for UNF can be found here: <https://rasmusfl.github.io/volunteer.html>.

University of Copenhagen - Department of Mathematical Sciences

Student representative in the Board of Education (*June 2022 - January 2025*)

Miscellaneous

Languages: Danish (mother tongue), English (fluent), German (basic).

Interests: Reading, in particular science fiction, horror and fantasy. Learning German, running, swimming, teaching mathematics, folk dance. Music (wide range of genres such as rock, metal, synth, jazz and soul among others).